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COM232

Assignment 3: Predicting Customer Purchase using Logistic Regression

An online store wants to predict whether a visitor will purchase a product ($y = 1$) or leave without buying ($y = 0$).

We collect two features about each customer:

- x_1 = Time spent browsing the website (minutes)
- x_2 = Number of product pages viewed
- y = Purchase outcome (1 = purchase, 0 = no purchase)

We need to use logistic regression to model the probability of purchase:

$$\hat{y} = \sigma(w_1x_1 + w_2x_2 + b)$$

where

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Dataset

Customer	Time on site (x_1)	Pages viewed (x_2)	Purchase (y)
A	1	4	0
B	2	3	0
C	3	7	1
D	5	2	1
E	6	6	1

Initial model parameters:

- $m_1 = 0.8$

- $m_2 = 0.4$

- $b = -4.0$

Tasks

1. Compute Probabilities (5 points)

For each customer A to E:

- Compute the linear score:

$$z = w_1x_1 + w_2x_2 + b$$

- Compute the predicted probability:

$$\hat{y} = \sigma(z)$$

Handwritten calculations for predicted probabilities for customers A through E:

CUST

A : $z = 0.8(1) + 0.4(4) - 4 = -1.6$, $\hat{y} = \sigma(-1.6) \approx 0.167981$

B : $z = 0.8(2) + 0.4(3) - 4 = -1.2$, $\hat{y} = \sigma(-1.2) \approx 0.231475$

C : $z = 0.8(3) + 0.4(7) - 4 = 1.2$, $\hat{y} = \sigma(1.2) \approx 0.768525$

D : $z = 0.8(5) + 0.4(2) - 4 = 0.8$, $\hat{y} = \sigma(0.8) \approx 0.689974$

E : $z = 0.8(6) + 0.4(6) - 4 = 3.2$, $\hat{y} = \sigma(3.2) \approx 0.960834$

Customer	Time on site (x_1)	Pages viewed (x_2)	Purchase (y)	$\hat{y}$
A	1	4	0	0.167981
B	2	3	0	0.231475
C	3	7	1	0.768525
D	5	2	1	0.689974
E	6	6	1	0.960834

2. Compute Average Loss (6 points)

Compute the Binary Cross-Entropy (BCE) loss for each sample:

$$Loss = -[y \ln(\hat{y}) + (1 - y) \ln(1 - \hat{y})]$$

- Fill in the table by reporting the **loss per customer**

Loss

$$A: \text{Loss} = -[0 \ln(0.167981) + (1-0) \ln(1-0.167981)] = 0.1839,$$

$$B: \text{Loss} = -[0 \ln(0.231475) + (1-0) \ln(1-0.231475)] = 0.2632,$$

$$C: \text{Loss} = -[1 \ln(0.768525) + (1-1) \ln(1-0.768525)] = 0.2632,$$

$$D: \text{Loss} = -[1 \ln(0.689974) + (1-1) \ln(1-0.689974)] = 0.3711,$$

$$E: \text{Loss} = -[1 \ln(0.960834) + (1-1) \ln(1-0.960834)] = 0.0399,$$

$$\text{Average Loss} = \frac{1}{5} (0.1839 + 0.2632 + 0.2632 + 0.3711 + 0.0399) \\ = 0.22426$$

Customer	Time on site (x_1)	Pages viewed (x_2)	Purchase (y)	$\hat{y}$	Loss
A	1	4	0	0.167981	0.1839
B	2	3	0	0.231475	0.2632
C	3	7	1	0.768525	0.2632
D	5	2	1	0.689974	0.3711
E	6	6	1	0.960834	0.0399

- Report the **average BCE loss** over all 5 customers

$$Loss_{avg} = \frac{1}{N} \sum_{i=1}^N Loss_i$$

$$Loss_{avg} = 0.22462$$

3. Update the slope and intercept using Gradient Descent (6 points)

Perform **one step of gradient descent** with learning rate $\eta = 0.1$:

DIFFERENCE OF $\hat{y}_i$ and y_i
LIST

$$A: r = 0.167081 - 0 = 0.167081 //$$

$$B: r = 0.231475 - 0 = 0.231475 //$$

$$C: r = 0.768525 - 1 = -0.231475 //$$

$$D: r = 0.680974 - 1 = -0.319026 //$$

$$E: r = 0.90834 - 1 = -0.09166 //$$

$$\begin{aligned} 1) \quad \frac{dL}{dm_1} &= \frac{1}{5} \sum r \cdot x_1 \\ &= \frac{1}{5} (A(1) + B(2) + C(3) + D(5) + E(6)) \\ \frac{dL}{dm_1} &= -0.369824 \end{aligned}$$

$$\begin{aligned} 2) \quad \frac{dL}{dm_2} &= \frac{1}{5} \sum r \cdot x_2 \\ &= \frac{1}{5} (A(4) + B(3) + C(7) + D(2) + E(6)) \end{aligned}$$

$$\frac{dL}{dm_2} = -0.221805 //$$

$$\begin{aligned} 3) \quad \frac{dL}{db} &= \frac{1}{5} \sum r \\ &= \frac{1}{5} (A + B + C + D + E) \end{aligned}$$

$$\frac{dL}{db} = -0.036242 //$$

- Compute gradients of the loss w.r.t. m_1, m_2, b

$$\frac{\partial L}{\partial m_1} = -0.036242, \quad \frac{\partial L}{\partial m_2} = -0.221805, \quad \frac{\partial L}{\partial b} = -0.036242$$

- Calculate the new slopes m_1, m_2 and intercept b :

UPDATE NEW SLOPES & b:

$$\begin{aligned} \text{new } m_1 &= m_1 - \eta \frac{\partial L}{\partial m_1} = 0.8 - 0.1(-0.036242) = 0.837 \\ \text{new } m_2 &= m_2 - \eta \frac{\partial L}{\partial m_2} = 0.4 - 0.1(-0.221805) = 0.422 \\ \text{new } b &= b - \eta \frac{\partial L}{\partial b} = -4.0 - 0.1(-0.036242) = -3.996 \end{aligned}$$

$$\text{new } m_1 = m_1 - \eta \frac{\partial L}{\partial m_1}, \quad \text{new } m_2 = m_2 - \eta \frac{\partial L}{\partial m_2}, \quad \text{new } b = b - \eta \frac{\partial L}{\partial b}$$

$$\text{new } m_1 = 0.837, \quad \text{new } m_2 = 0.422, \quad \text{new } b = -3.996$$

- Compute new probabilities using the new slopes and intercept (5 points)

A) COMPUTE NEW PROBABILITIES USING NEW SLOPES AND INTERCEPT

COST

$$\begin{aligned} \text{A: } z &= 0.837(1) + 0.422(4) - 3.996 = -1.179, \quad \hat{y}_{\text{new}} = \sigma(-1.179) \approx 0.232 \\ \text{B: } z &= 0.837(2) + 0.422(5) - 3.996 = -1.0, \quad \hat{y}_{\text{new}} = \sigma(-1.0) \approx 0.2689 \\ \text{C: } z &= 0.837(3) + 0.422(7) - 3.996 = 1.5, \quad \hat{y}_{\text{new}} = \sigma(1.5) \approx 0.8181 \\ \text{D: } z &= 0.837(5) + 0.422(2) - 3.996 = 1.0, \quad \hat{y}_{\text{new}} = \sigma(1.0) \approx 0.7376 \\ \text{E: } z &= 0.837(6) + 0.422(6) - 3.996 = 3.6, \quad \hat{y}_{\text{new}} = \sigma(3.6) \approx 0.9745 \end{aligned}$$

Customer	Time on site (x_1)	Pages viewed (x_2)	Purchase (y)	new $\hat{y}$
A	1	4	0	0.1816882
B	2	3	0	0.258900
C	3	7	1	0.813118
D	5	2	1	0.737629
E	6	6	1	0.972451

5. Compute new Average Loss (6 points)

Fill in the table by computing the **new loss per customer**

WST

$$A: -[0 \ln(0.1816882) + (1-0) \ln(1-0.1816882)] = 0.2069$$

$$B: -[0 \ln(0.258900) + (1-0) \ln(1-0.258900)] = 0.2996$$

$$C: -[1 \ln(0.813118) + (1-1) \ln(1-0.813118)] = 0.2069$$

$$D: -[1 \ln(0.737629) + (1-1) \ln(1-0.737629)] = 0.3043$$

$$E: -[1 \ln(0.972451) + (1-1) \ln(1-0.972451)] = 0.027$$

$$\text{LOSS AVERAGE} = \frac{1}{5} (A+B+C+D+E)$$

$$\approx 0.209$$

Customer	Time on site (x_1)	Pages viewed (x_2)	Purchase (y)	$\hat{y}$	new Loss
A	1	4	0	0.1816882	0.2069
B	2	3	0	0.258900	0.2996
C	3	7	1	0.813118	0.2069
D	5	2	1	0.737629	0.3043
E	6	6	1	0.972451	0.027

- Report the **new average BCE loss** over all 5 customers. (Note: The new average loss must be lower than the previous average loss)

$$NewLoss_{avg} = 0.209$$